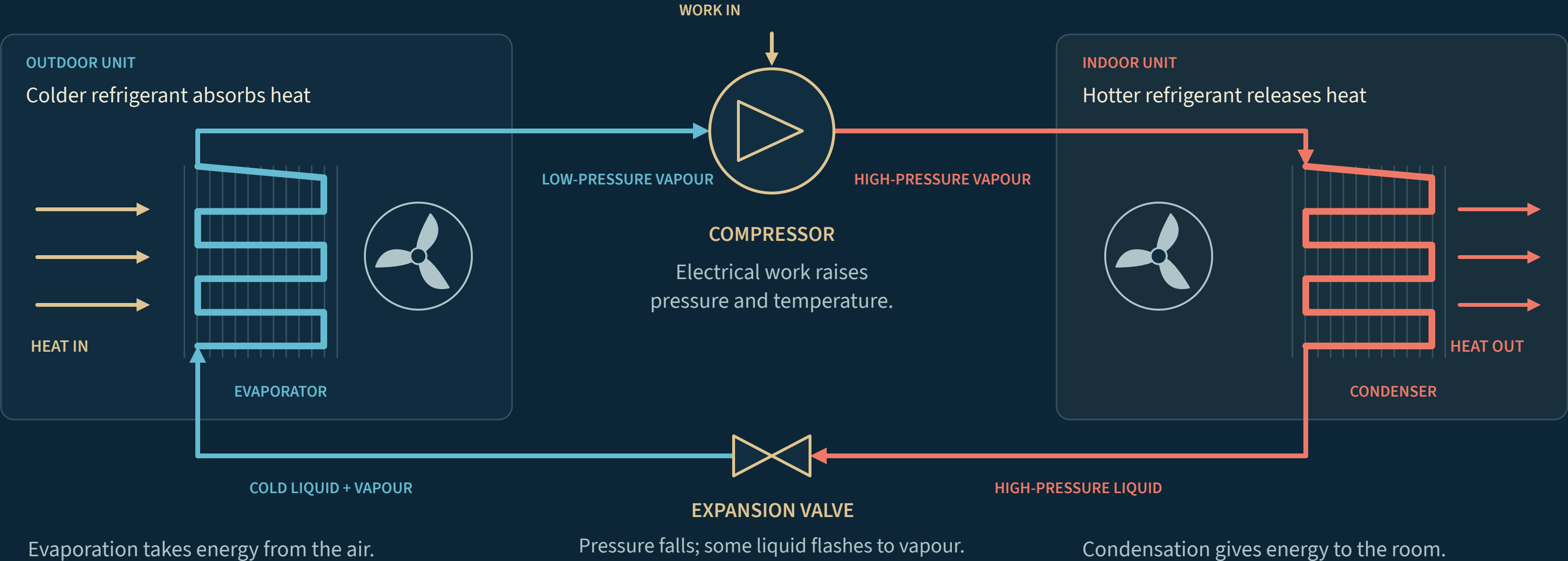


How a heat pump warms a room

A phase-changing fluid moves heat uphill in temperature. Electricity drives the transfer; feedback controls the demand.

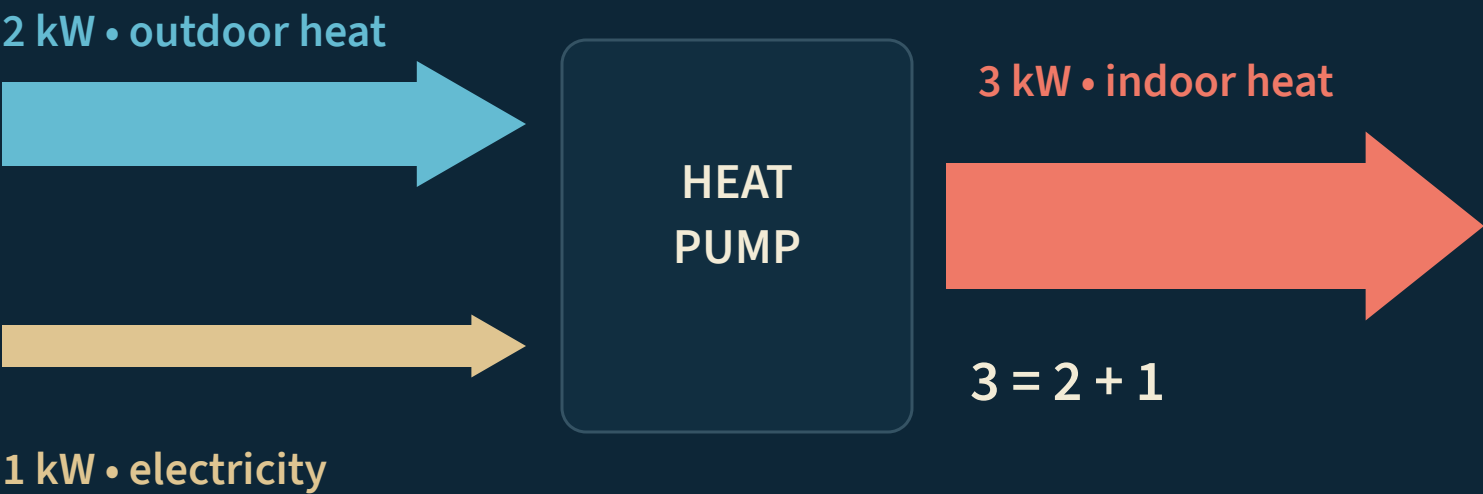
01 Follow the refrigerant, not the air

Heating mode • the closed circuit carries heat from outdoor air to the room.



02 More heat than electrical input

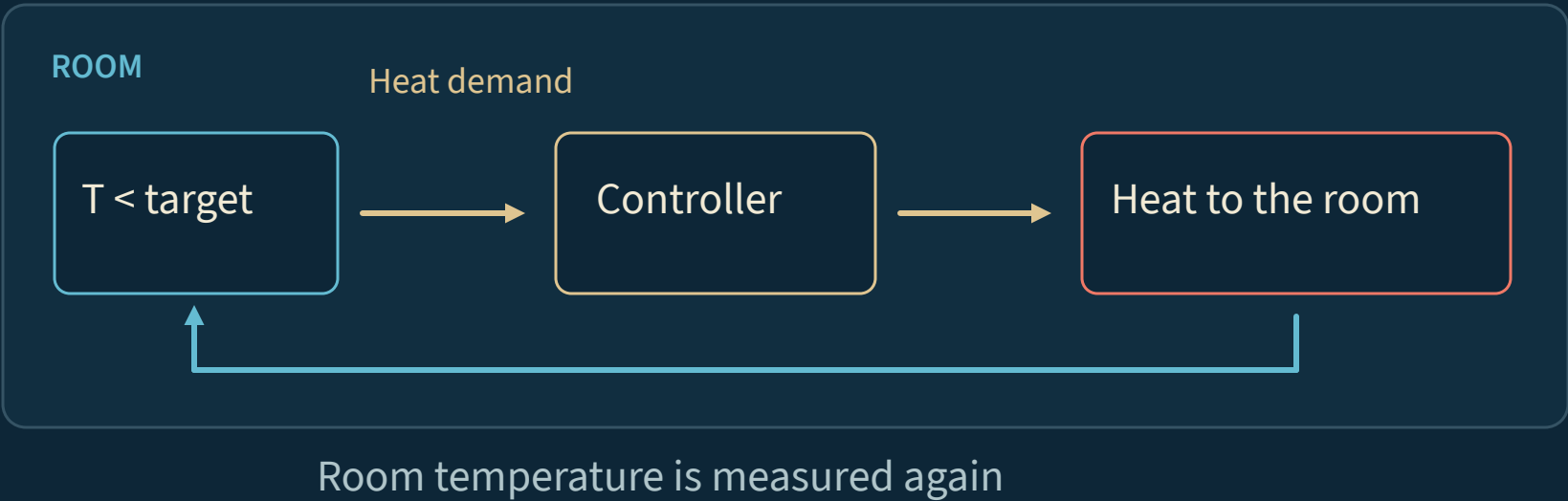
A worked balance • illustrative operating point, not a product rating.



COP = delivered heat / electrical input = 3

03 The thermostat closes another loop

Refrigerant transfers heat; the control signal regulates demand.



Real control may modulate output or cycle equipment.
Defrost and reversing valves are omitted from the circuit.